ECE 5254 - Homework #3

Buck Converter _ Voltage Mode Control

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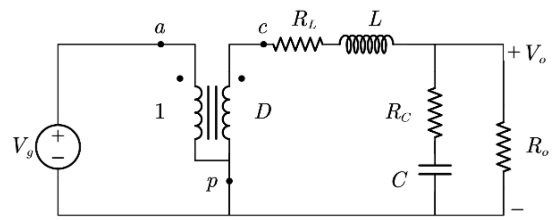
1) The small signal model using three-terminal switch is shown below.



The open loop output impedance transfer function can be derived as

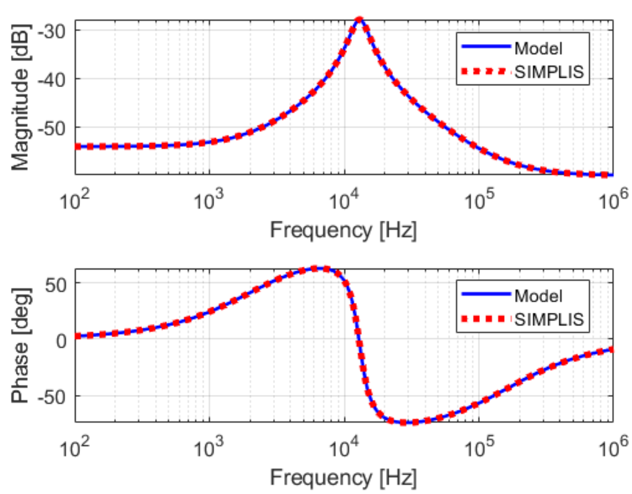
$$Z_o = R_o||(R_c + \frac{1}{sC_o})||(R_L + sL)$$

$$Z_o = \frac{R_o(R_L + sL)(sC_oR_o + 1)}{s^2C_oL(R_o + R_c) + s(C_oR_oR_L + C_oR_cR_o + C_oR_cR_L + L) + R_0 + R_L}$$

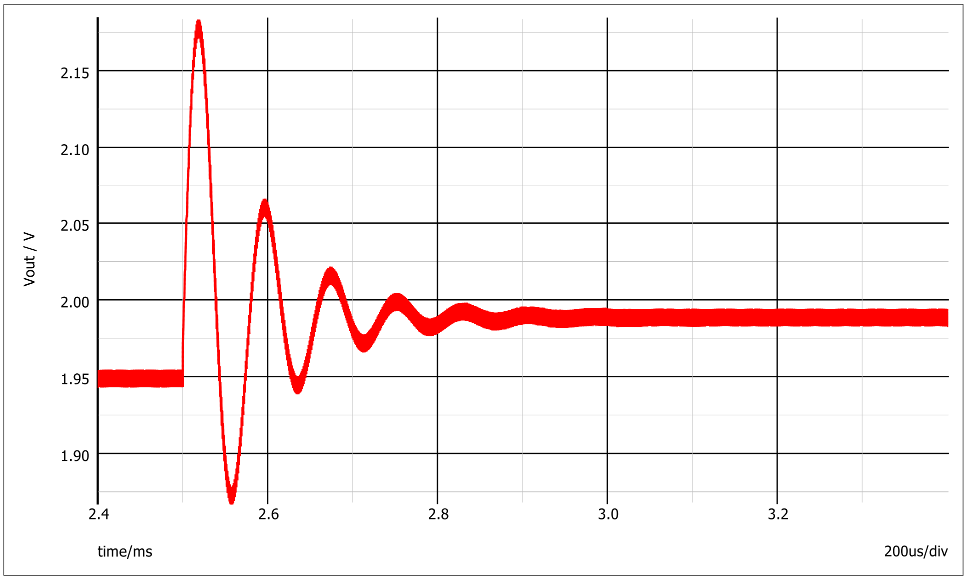
Using given values,

$$Z_o = \frac{0.000995s^2 + 1008.29187s + 13266998.3416252}{s^2 + 24941.95688s + 6699834162.52}$$

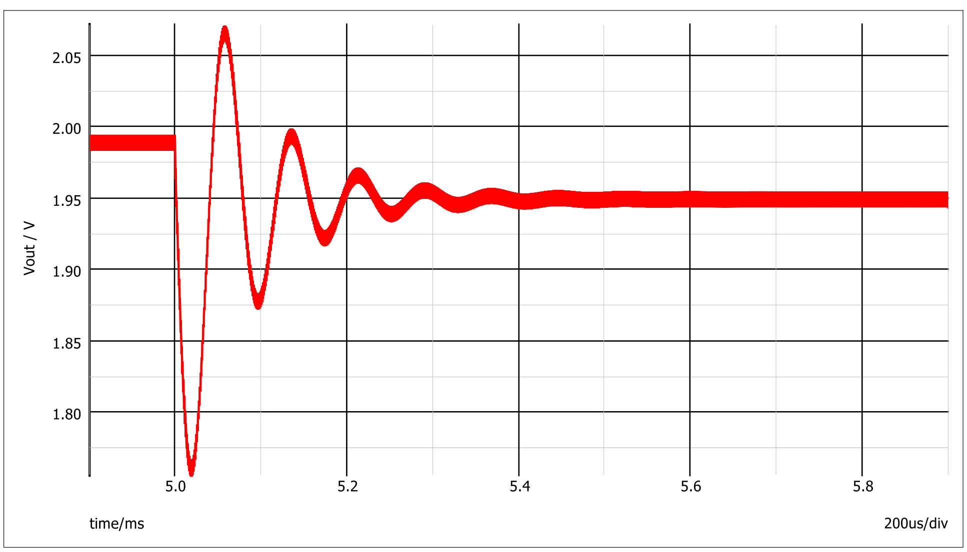
The open loop output impedance bode plot of model and simulation are shown below.



At the time load current steps down, the overshoot voltage is 228mV, with a settle time of 330us.



At the time load current steps up, the undershoot voltage is 226mV, with a settle time of 334us.



2). a.

Circuit parameters:

$R_O := 0.2$

$C := 1 \cdot 10^{-3}$ $\quad R_C := 0.001$

$L := 150 \cdot 10^{-9}$ $\quad R_L := 0.002$ $\quad V1 := 12$

Define:

$$w_0 := \frac{1}{\sqrt{L \cdot C \cdot \frac{R_O + R_C}{R_O + R_L}}} \rightarrow 81852.514698821133316$$

$$Q := \frac{1}{w_0} \cdot \frac{1}{\frac{L}{R_O + R_L} + C \cdot \left(R_C + \frac{R_O \cdot R_L}{R_O + R_L}\right)} \rightarrow 3.2817198379912994275$$

$$\Delta := 1 + \frac{1}{Q} \cdot \frac{s}{w_0} + \frac{s^2}{{w_0}^2} \rightarrow 0.14925742574257425743 \cdot 10^{-9} \cdot s^2 + 0.000003722772277227722723 \cdot s + 1.0$$

We have open loop output impedance and open loop output to control voltage transfer function as below:

$$G_{BUCK\_Z}(s) := \frac{R_O \cdot R_L}{R_O + R_L} \cdot \frac{(R_C \cdot C \cdot s + 1) \cdot \left(1 + s \cdot \frac{L}{R_L}\right)}{\Delta} \xrightarrow{float, 3} \frac{0.133 \cdot 10^8 \cdot (0.000001 \cdot s + 1.0) \cdot (0.000075 \cdot s + 1.0)}{s^2 + 0.249 \cdot 10^5 \cdot s + 0.67 \cdot 10^{10}}$$

$$G_{BUCK\_VD}(s) := V1 \cdot \frac{R_O}{R_O + R_L} \cdot \frac{R_C \cdot C \cdot s + 1}{\Delta} \xrightarrow{float, 3} \frac{0.796 \cdot 10^5 \cdot s + 0.796 \cdot 10^{11}}{s^2 + 0.249 \cdot 10^5 \cdot s + 0.67 \cdot 10^{10}}$$

Then we can design a type III compensator.

ESR zero:

$$w_Z := \frac{1}{R_C \cdot C} = 1 \cdot 10^6$$

Double pole resonant frequency:

$$w_0 \rightarrow 81852.514698821133316 = 8.185 \cdot 10^4$$

Cross over frequency(BW):

$fc := 80 \cdot 10^3$ $\quad w_C := 2 \cdot \pi \cdot fc = 5.027 \cdot 10^5$

The first pole in compensator are designed to cancel ESR zero:

$$w_{P1} := w_Z = 1 \cdot 10^6$$

The second pole in compensator are set to ten times the cross over frequency:

$$w_{P2} := 10 \cdot w_C = 5.027 \cdot 10^6$$

Place two compensator zeroes below cross over frequency and on both sides of the double poles.

$$w_{Z1} := 16 \cdot 10^3$$

$$w_{Z2} := 100 \cdot 10^3$$

To calculate the integrator coeffcient, first we define:

$$B(s) := \frac{1+\frac{s}{w_{Z1}}}{1+\frac{s}{w_{P1}}} \cdot \frac{1+\frac{s}{w_{Z2}}}{1+\frac{s}{w_{P2}}} \rightarrow \frac{(5026548.24574367 \cdot 0.00001 \cdot s + 5026548.24574367) \cdot (0.0000625 \cdot s + 1.0)}{(s + 5026548.24574367) \cdot (0.000001 \cdot s + 1.0)}$$

Thus,

$$w_1 := \frac{w_C}{\left|B(1\mathrm{j} \cdot w_C) \cdot G_{BUCK\_VD}(1\mathrm{j} \cdot w_C)\right|} = 9.702 \cdot 10^3$$

Then we have compensator, loop gain, and closed loop output impedance derives as below:

$$A(s) := -\frac{w_1}{s} \cdot \frac{1+\frac{s}{w_{Z1}}}{1+\frac{s}{w_{P1}}} \cdot \frac{1+\frac{s}{w_{Z2}}}{1+\frac{s}{w_{P2}}} \xrightarrow{simplify,\, float,\, 3} \frac{-(0.305 \cdot 10^8) \cdot s^2 - 0.354 \cdot 10^{13} \cdot s - 0.488 \cdot 10^{17}}{s \cdot (s + 0.1 \cdot 10^7) \cdot (s + 0.503 \cdot 10^7)}$$

$$T(s) := -G_{BUCK\_VD}(s) \cdot A(s) \xrightarrow{float,\, 3} \frac{(-(0.796 \cdot 10^5) \cdot s - 0.796 \cdot 10^{11}) \cdot (-(0.305 \cdot 10^8) \cdot s^2 - 0.354 \cdot 10^{13} \cdot s - 0.488 \cdot 10^{17})}{s \cdot (s + 0.1 \cdot 10^7) \cdot (s + 0.503 \cdot 10^7) \cdot (s^2 + 0.249 \cdot 10^5 \cdot s + 0.67 \cdot 10^{10})}$$

$$Z(s) := \frac{G_{BUCK\_Z}(s)}{1+T(s)} \xrightarrow{factor,\, float,\, 3} \frac{0.000998 \cdot s \cdot (s + 0.503 \cdot 10^7) \cdot (s^2 + 0.249 \cdot 10^5 \cdot s + 0.67 \cdot 10^{10}) \cdot (s + 0.133 \cdot 10^5) \cdot (s + 0.1 \cdot 10^7)}{(s^4 + 0.505 \cdot 10^7 \cdot s^3 + 0.256 \cdot 10^{13} \cdot s^2 + 0.315 \cdot 10^{18} \cdot s + 0.388 \cdot 10^{22}) \cdot (s^2 + 0.249 \cdot 10^5 \cdot s + 0.67 \cdot 10}$$

Verify the cross over frequency and phase margin:

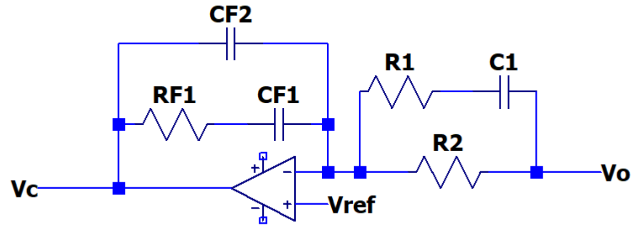
At 80kHz, loop gain is:

$$20 \cdot \log(|T(2\mathrm{j} \cdot \pi \cdot fc)|) = -2.235 \cdot 10^{-4}$$

Phase margin:

$$\arg(T(2\mathrm{j} \cdot \pi \cdot fc)) + 180° = 74.124\ deg$$

According to the compensator transfer frequency, the components values can be calculated.



First, define value of R1.

$R1 := 10 \cdot 10^{3}$

The others can be calculated.

$$C1 := \frac{1}{w_{P1} \cdot R1} = 1 \cdot 10^{-10}$$

$$R2 := \frac{1}{w_{Z2} \cdot C1} = 1 \cdot 10^{5}$$

$$CF2 := \frac{w_{Z1}}{w_{P2} \cdot R2 \cdot w_{1}} = 3.281 \cdot 10^{-12}$$

$$CF1 := \frac{1}{w_{1} \cdot R2} = 1.031 \cdot 10^{-9}$$

$$RF1 := \frac{1}{w_{Z1} \cdot CF1} = 6.064 \cdot 10^{4}$$

2). b.

The results are verified with Simplis. The diagrams are shwon below.

$f := \mathrm{logspace}(100, 1000000, 1000)$

A1 := READEXCEL (".\2bcomp.xlsx", "1!A2:C402")

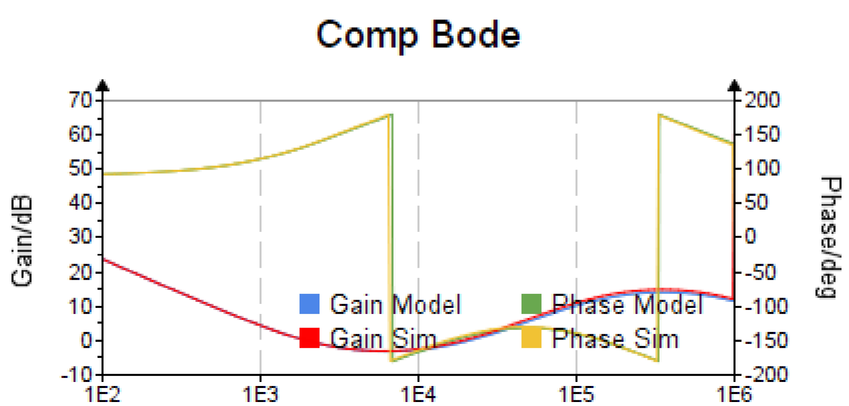
$$\mathrm{A1} = \begin{bmatrix} 100 & \\ 102.329 & \\ 104.713 & \\ 107.152 & \\ & \ddots \end{bmatrix} \quad FREQ := \mathrm{A1}^{\langle 0 \rangle} = \begin{bmatrix} 100 \\ 102.329 \\ 104.713 \\ 107.152 \\ \vdots \end{bmatrix} \quad GAIN := \mathrm{A1}^{\langle 1 \rangle} = \begin{bmatrix} 23.759 \\ 23.559 \\ 23.359 \\ 23.16 \\ \vdots \end{bmatrix} \quad PHASE := \mathrm{A1}^{\langle 2 \rangle} = \begin{bmatrix} 92.6 \\ 92.661 \\ 92.723 \\ 92.786 \\ \vdots \end{bmatrix}$$

Inputs

$\mathbf{X}_1 := f$ $\qquad$ $\mathbf{X}_2 := f$

$\mathbf{Y}_1 := \overrightarrow{20 \cdot \log(|A(2\mathrm{j} \cdot \pi \cdot f)|)}$ $\qquad$ $\mathbf{Y}_2 := \overrightarrow{\arg(A(2\mathrm{j} \cdot \pi \cdot f))} \dfrac{180}{\pi}$

$\mathbf{X}_3 := FREQ \quad \mathbf{Y}_3 := GAIN \quad \mathbf{X}_4 := FREQ \quad \mathbf{Y}_4 := PHASE$



$A1 := \mathrm{READEXCEL}(\text{“.\textbackslash 2bloop.xlsx”}, \text{“1!A2:C402”})$

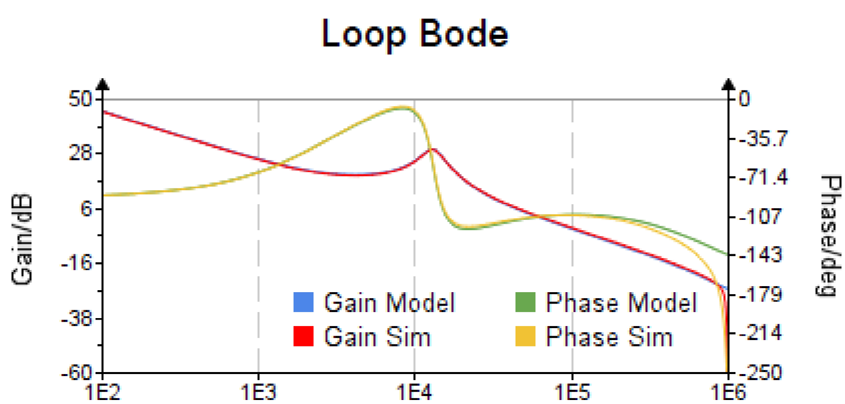
$$A1 = \begin{bmatrix} 100 \\ 102.329 \\ 104.713 \\ 107.152 \\ \ddots \end{bmatrix} \quad FREQ := A1^{\langle 0 \rangle} = \begin{bmatrix} 100 \\ 102.329 \\ 104.713 \\ 107.152 \\ \vdots \end{bmatrix} \quad GAIN := A1^{\langle 1 \rangle} = \begin{bmatrix} 45.109 \\ 44.91 \\ 44.71 \\ 44.51 \\ \vdots \end{bmatrix} \quad PHASE := A1^{\langle 2 \rangle} = \begin{bmatrix} -87.565 \\ -87.509 \\ -87.451 \\ -87.391 \\ \vdots \end{bmatrix}$$

Inputs

$\mathbf{X}_1 := f$ $\qquad$ $\mathbf{X}_2 := f$

$\mathbf{Y}_1 := \overrightarrow{20 \cdot \log(|T(2j \cdot \pi \cdot f)|)}$ $\qquad$ $\mathbf{Y}_2 := \overrightarrow{\arg(T(2j \cdot \pi \cdot f))} \dfrac{180}{\pi}$

$\mathbf{X}_3 := FREQ \quad \mathbf{Y}_3 := GAIN \quad \mathbf{X}_4 := FREQ \quad \mathbf{Y}_4 := PHASE$



Model:

phase margin: 74.124 °

Simulation:

phase margin: 74°

gain margin: 24dB

$A1 := \mathrm{READEXCEL}(\text{“.\textbackslash 2bZ.xlsx”}, \text{“1!A2:C402”})$

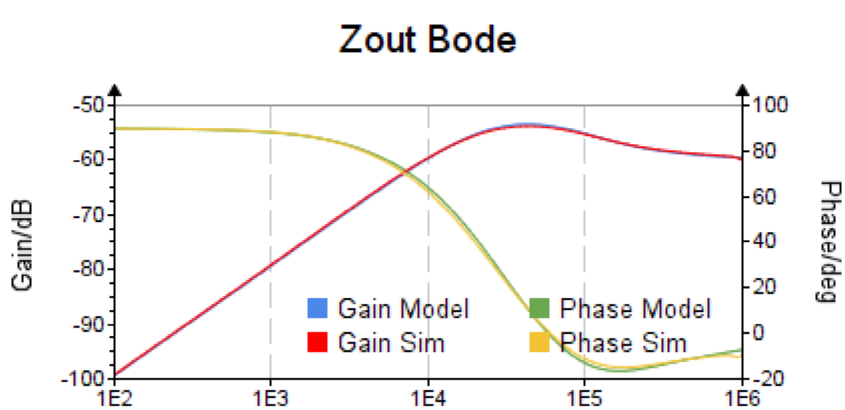
$$A1 = \begin{bmatrix} 100 \\ 102.329 \\ 104.713 \\ 107.152 \\ \ddots \end{bmatrix} \quad FREQ := A1^{\langle 0 \rangle} = \begin{bmatrix} 100 \\ 102.329 \\ 104.713 \\ 107.152 \\ \vdots \end{bmatrix} \quad GAIN := A1^{\langle 1 \rangle} = \begin{bmatrix} -99.166 \\ -98.966 \\ -98.766 \\ -98.566 \\ \vdots \end{bmatrix} \quad PHASE := A1^{\langle 2 \rangle} = \begin{bmatrix} 89.845 \\ 89.841 \\ 89.838 \\ 89.834 \\ \vdots \end{bmatrix}$$

Inputs

$\mathbf{X}_1 := f$ $\qquad$ $\mathbf{X}_2 := f$

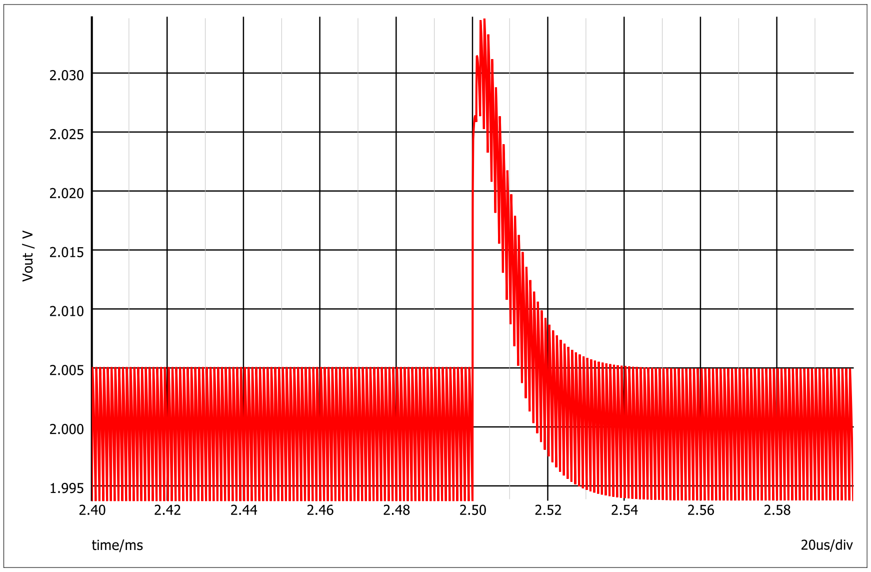
$\mathbf{Y}_1 := \overrightarrow{20 \cdot \log(|Z(2j \cdot \pi \cdot f)|)}$ $\qquad$ $\mathbf{Y}_2 := \overrightarrow{\arg(Z(2j \cdot \pi \cdot f))} \dfrac{180}{\pi}$

$\mathbf{X}_3 := FREQ \quad \mathbf{Y}_3 := GAIN \quad \mathbf{X}_4 := FREQ \quad \mathbf{Y}_4 := PHASE$

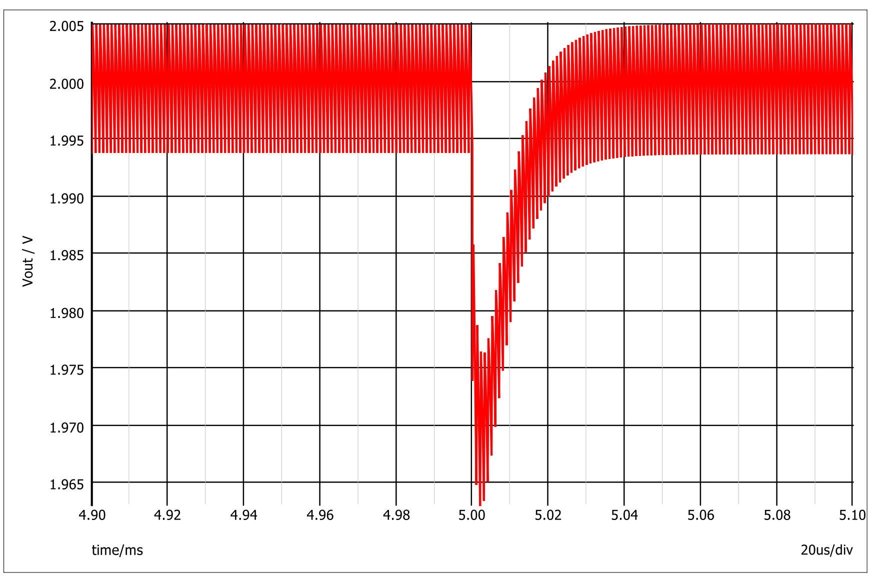


2) c.

At the time load current steps down, the overshoot voltage is 30mV, with a settling time less than 40us.



At the time load current steps up, the undershoot voltage is 30mV, with a settling time less than 40us.



Compared with open loop control, closed loop control reduces the settling time and suppresses the overshoot/under shoot voltage. It ensures a more stabilized and faster load transient performance, along with very low DC error.

2) d.

After adjusting the first zero frequency to 1/5 of its original value, the integrator gain w1 and components' values are calculated in MATHCAD as below.

$$R1 := 10 \cdot 10^{3}$$

$$C1 := \frac{1}{w_{P1} \cdot R1} = 1 \cdot 10^{-10}$$

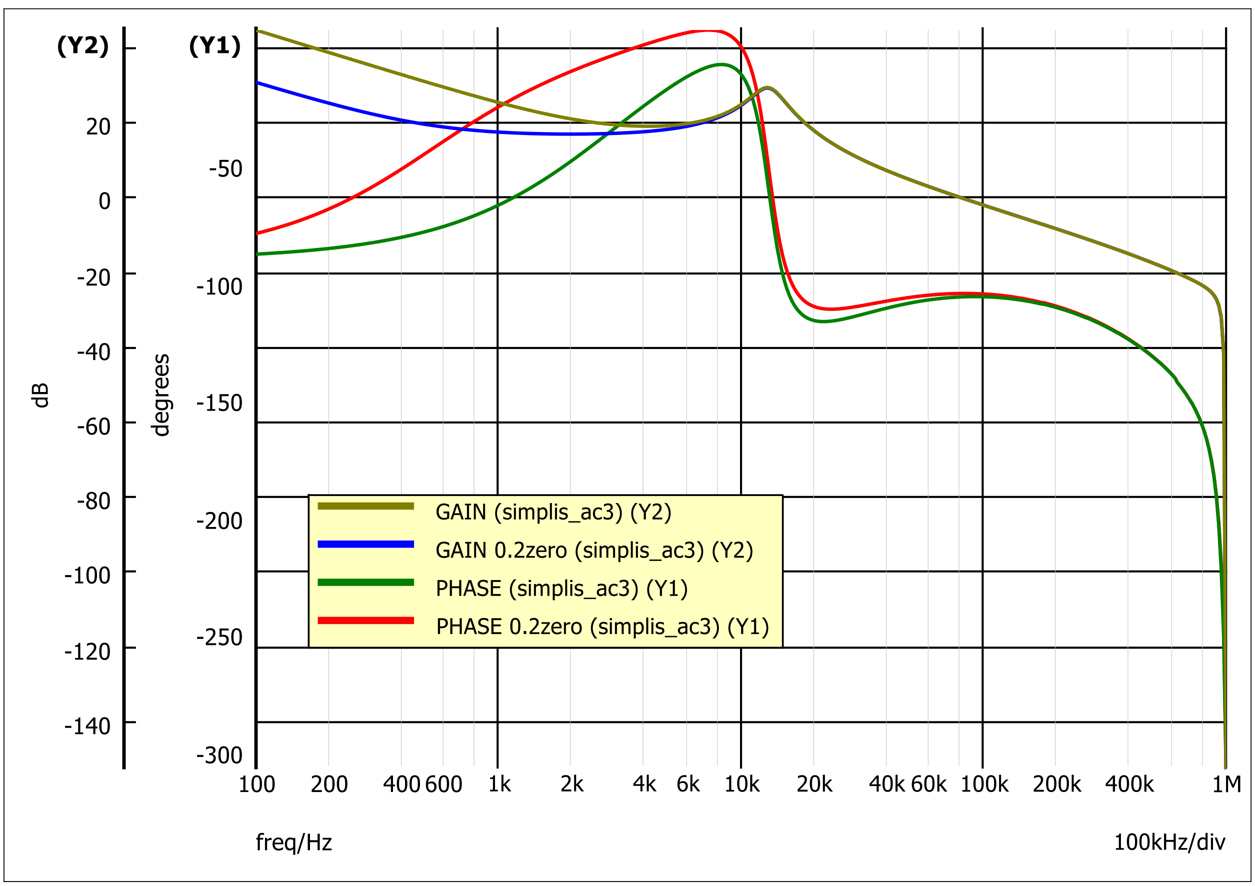
$$R2 := \frac{1}{w_{Z2} \cdot C1} = 1 \cdot 10^{5}$$

$$CF2 := \frac{w_{Z1}}{w_{P2} \cdot R2 \cdot w_{1}} = 3.279 \cdot 10^{-12}$$

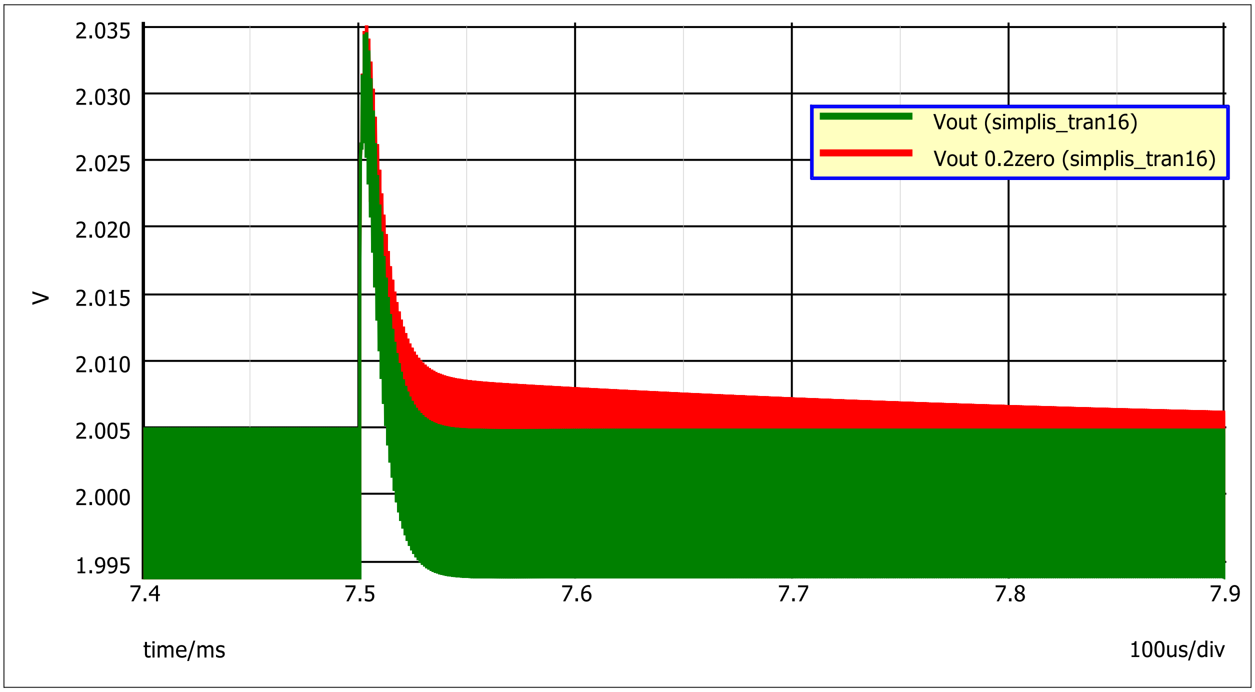
$$CF1 := \frac{1}{w_{1} \cdot R2} = 5.151 \cdot 10^{-9}$$

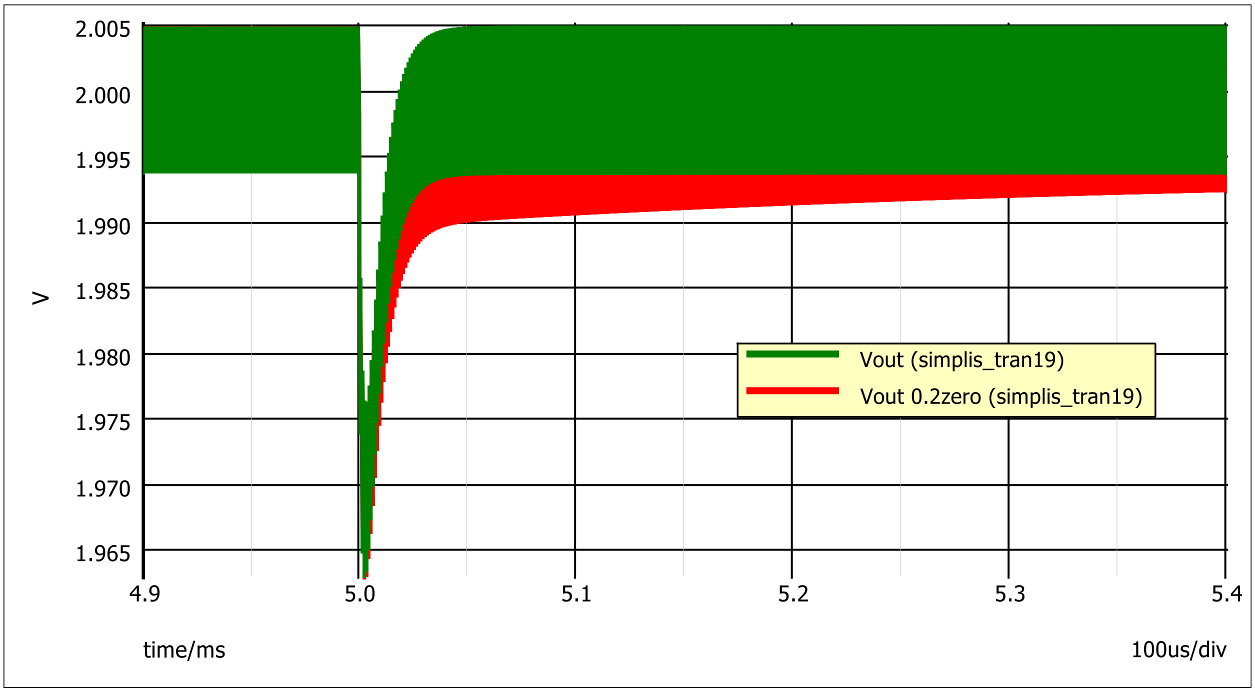
$$RF1 := \frac{1}{w_{Z1} \cdot CF1} = 6.067 \cdot 10^{4}$$

The bode plots of loop gain in two different cases are simulated and shown in one diagram below. We can observe that the gain at low frequency decreases and phase is boosted.



The transient performance at load step down/ step up in two cases are compared in two diagrams, respectively.





We can observe with a lower frequency compensator zero, the settling time becomes longer. That is resulted by the transformation of compensator zero to loop pole.

3) a.

The zeros and poles of compensators and components' values are calculated in MATHCAD.

For 200kHz bandwidth, the high frequency pole is adjusted to not go beyond switching frequency, w1 is adjusted accordingly. Exact values are shown below.

$$fc := 200 \cdot 10^{3} \qquad w_C := 2 \cdot \pi \cdot fc = 1.257 \cdot 10^{6}$$

$$w_{P1} := w_Z = 1 \cdot 10^{6}$$

$$w_{P2} := 4 \cdot w_C = 5.027 \cdot 10^{6}$$

$$w_{Z1} := 16 \cdot 10^{3}$$

$$w_{Z2} := 100 \cdot 10^{3}$$

$$R1 := 10 \cdot 10^{3}$$

$$C1 := \frac{1}{w_{P1} \cdot R1} = 1 \cdot 10^{-10}$$

$$R2 := \frac{1}{w_{Z2} \cdot C1} = 1 \cdot 10^{5}$$

$$CF2 := \frac{w_{Z1}}{w_{P2} \cdot R2 \cdot w_1} = 1.232 \cdot 10^{-12}$$

$$CF1 := \frac{1}{w_1 \cdot R2} = 3.869 \cdot 10^{-10}$$

$$RF1 := \frac{1}{w_{Z1} \cdot CF1} = 1.615 \cdot 10^{5}$$

For 80kHz bandwidth, the compensator is designed in 2). a. Exact values are shown below.

$$fc := 80 \cdot 10^{3} \qquad w_C := 2 \cdot \pi \cdot fc = 5.027 \cdot 10^{5}$$

$$w_{P1} := w_Z = 1 \cdot 10^{6}$$

$$w_{P2} := 10 \cdot w_C = 5.027 \cdot 10^{6}$$

$$w_{Z1} := 16 \cdot 10^{3}$$

$$w_{Z2} := 100 \cdot 10^{3}$$

$$R1 := 10 \cdot 10^{3}$$

$$C1 := \frac{1}{w_{P1} \cdot R1} = 1 \cdot 10^{-10}$$

$$R2 := \frac{1}{w_{Z2} \cdot C1} = 1 \cdot 10^{5}$$

$$CF2 := \frac{w_{Z1}}{w_{P2} \cdot R2 \cdot w_1} = 3.281 \cdot 10^{-12}$$

$$CF1 := \frac{1}{w_1 \cdot R2} = 1.031 \cdot 10^{-9}$$

$$RF1 := \frac{1}{w_{Z1} \cdot CF1} = 6.064 \cdot 10^{4}$$

For 1kHz bandwidth, the high frequency pole is adjusted to higher times of cross over frequency, w1 is adjusted accordingly. Exact values are shown below.

$$fc := 1 \cdot 10^{3} \qquad w_C := 2 \cdot \pi \cdot fc = 6.283 \cdot 10^{3}$$

$$w_{P1} := w_Z = 1 \cdot 10^{6}$$

$$w_{P2} := 100 \cdot w_C = 6.283 \cdot 10^{5}$$

$$w_{Z1} := 16 \cdot 10^{3}$$

$$w_{Z2} := 100 \cdot 10^{3}$$

$$R1 := 10 \cdot 10^{3}$$

$$C1 := \frac{1}{w_{P1} \cdot R1} = 1 \cdot 10^{-10}$$

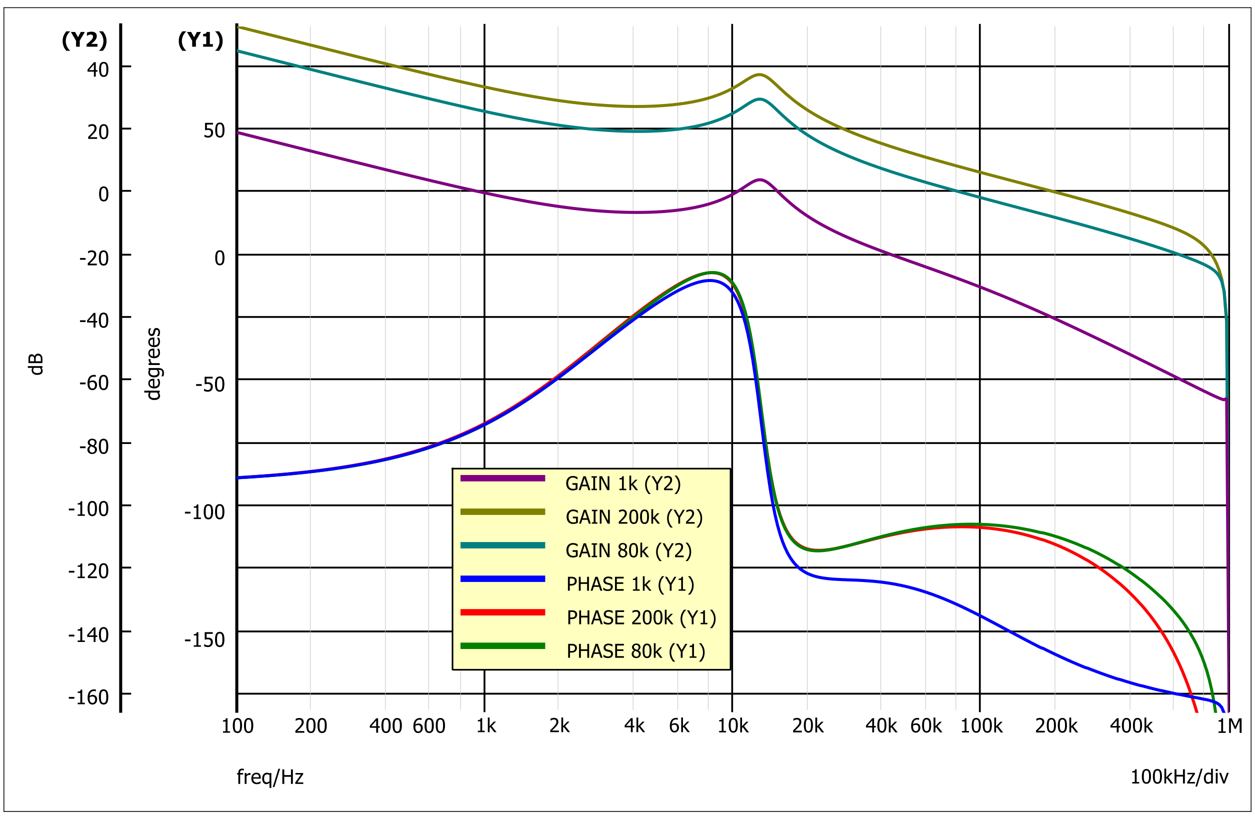
$$R2 := \frac{1}{w_{Z2} \cdot C1} = 1 \cdot 10^{5}$$

$$CF2 := \frac{w_{Z1}}{w_{P2} \cdot R2 \cdot w_1} = 5.212 \cdot 10^{-10}$$

$$CF1 := \frac{1}{w_1 \cdot R2} = 2.047 \cdot 10^{-8}$$

$$RF1 := \frac{1}{w_{Z1} \cdot CF1} = 3.053 \cdot 10^{3}$$

The bode plots of loop gain in three cases are simulated and shown in

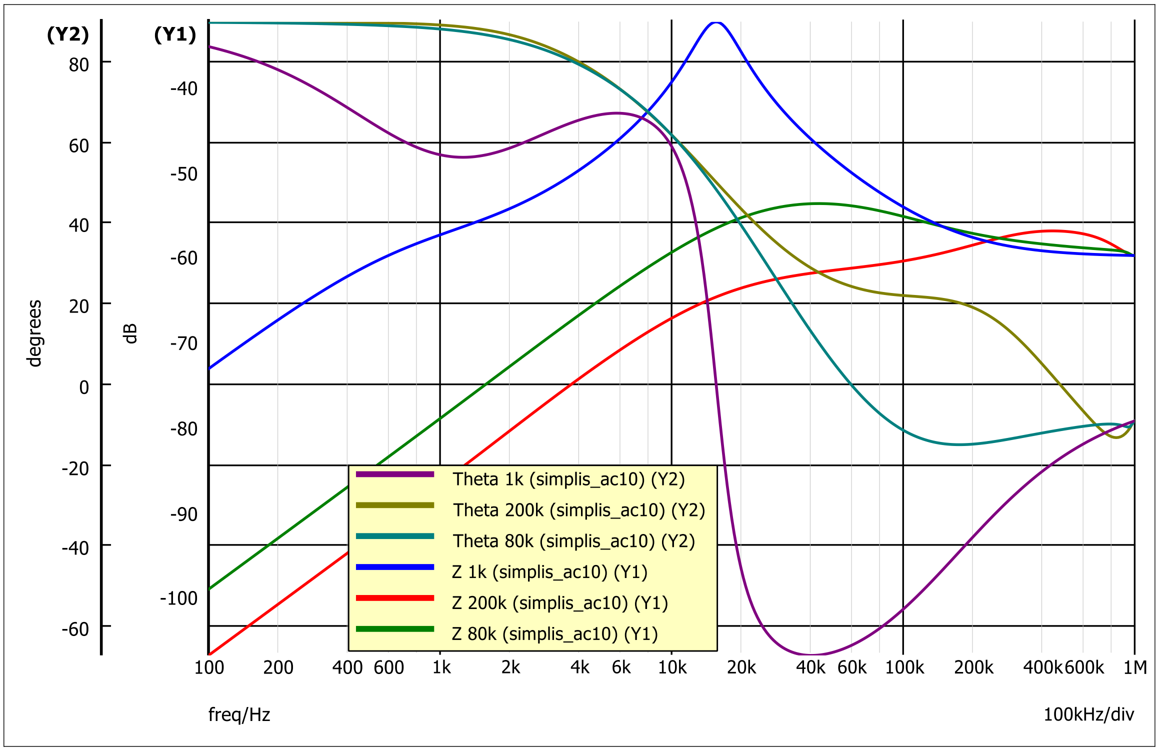


The phase margin and gain margin are concluded as follows.

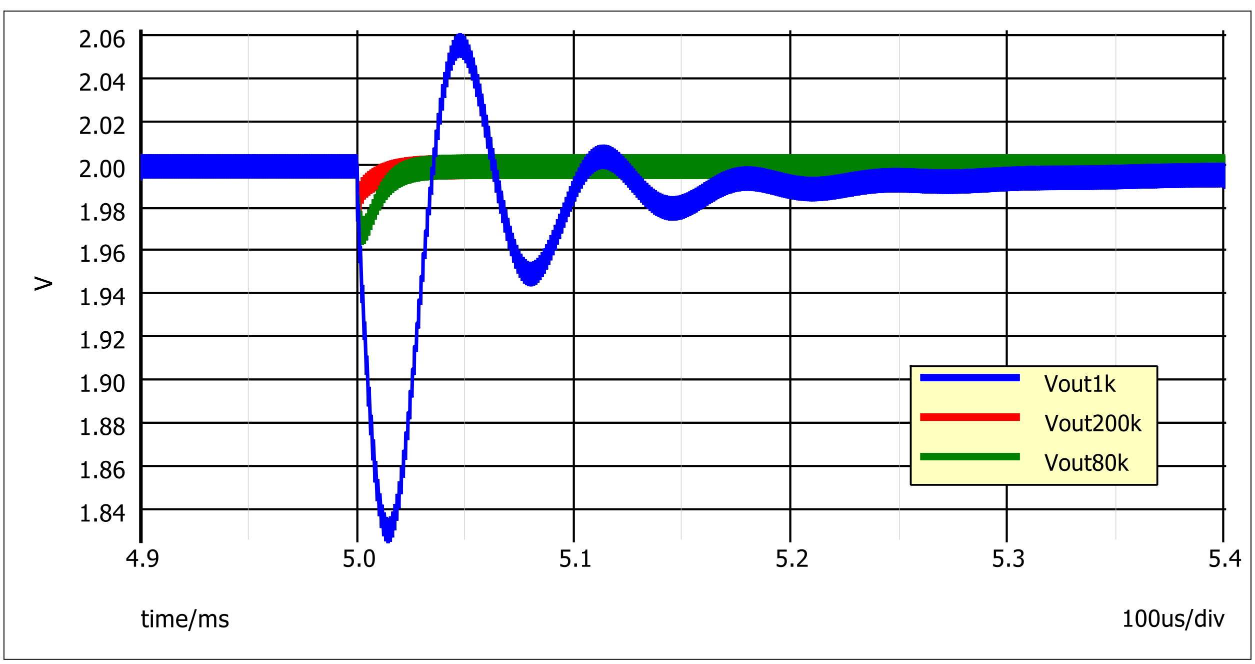
	1kHz	80kHz	200kHz
Phase margin /°	111	74	66
Gain margin /dB	65	24	13.6

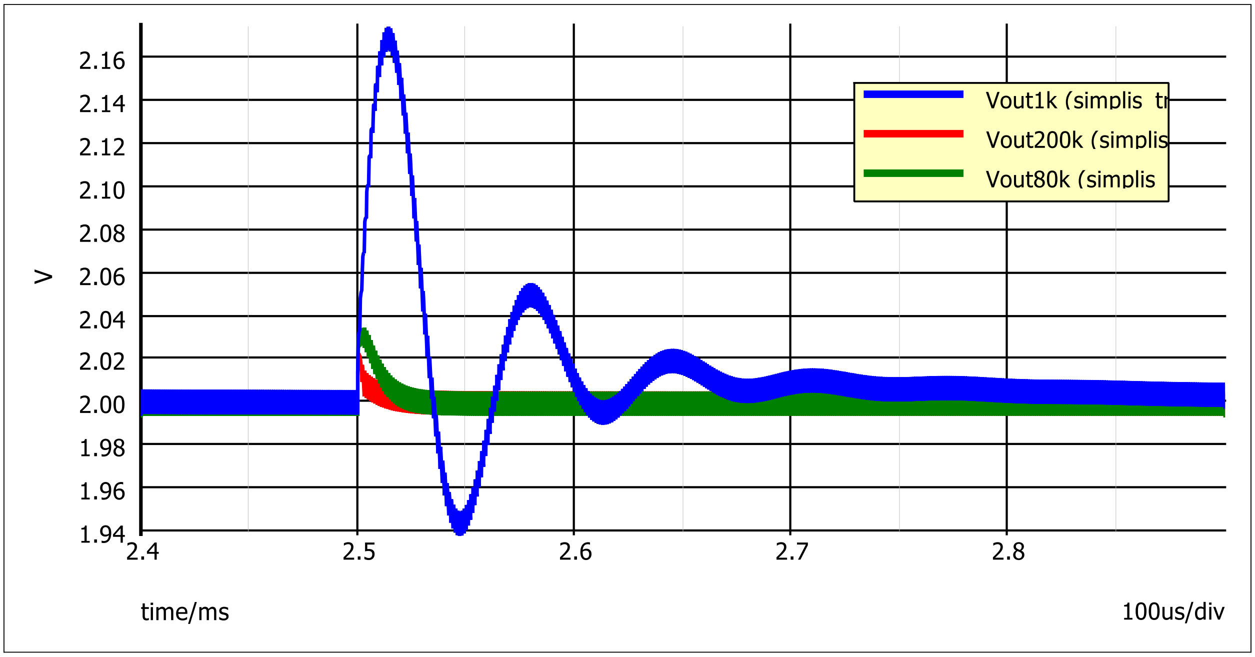
3) b.

The output impedances of three cases are simulated and shown in diagram below.



The load transient performances at the time load step up/down in three cases are simulated and shown in diagrams below, respectively.





We can observe a peak around 15kHz in case of 1kHz bandwidth, from the output impedance curve. In time domain, it is represented by an oscillation with a period of approximate 70us, as we see in the transient waveform. From the diagrams above, higher control bandwidth will lead to a lower overshoot/undershoot value and shorter settling time.

3) c.

There is no need to push control frequency to extreme high since:
First, in most cases, bandwidth will not go beyond the switching frequency.
Second, higher bandwidth will introduce higher noise, that means there will be higher possibility for a control failure. Also, control bandwidth is limited by sensor bandwidth. Higer sensor bandwidth always means higher price and extra complexity.
Third, in high frequency, the overshoot and undershoot are determined by output capacitor impedance. This impedance, also called PDN, is designed by choosing proper passive capacitors and optimizing structure and layout, but not controlled by closed loops.